

If the line P_1P_2 is contained in P_3 , then there are infinitely many solutions (any point on the line is a solution).

If the line P_1P_2 is parallel to P_3 (and does not intersect), then there is no solution.

Recall:

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$$

$$A \text{ is invertible} \iff \det(A) \neq 0$$

Start with the homogeneous case:

$$AX = 0$$

Example:

$$\begin{cases} x + z = 0 \\ x + y = 0 \\ x + 2y + 3z = 0 \end{cases}$$

There is always an obvious solution:

$$(0, 0, 0) \quad (\text{trivial solution})$$

The origin is always a solution because the three planes pass through the origin.

If $\det(A) \neq 0$, we can invert A :

$$AX = 0 \iff X = A^{-1}0 = 0$$

If $\det(A) = 0$

$$\iff \det(\vec{N}_1, \vec{N}_2, \vec{N}_3) = 0$$

Then $\vec{N}_1, \vec{N}_2, \vec{N}_3$ are coplanar.
(Parallelepiped has volume 0)

The line through the origin perpendicular to the plane spanned by $\vec{N}_1, \vec{N}_2, \vec{N}_3$ is parallel to all three planes and in fact contained in them.

$\Rightarrow$ Infinitely many solutions

Example:

$$\vec{N}_1 \times \vec{N}_2 \perp \vec{N}_1 \quad \text{and} \quad \vec{N}_1 \times \vec{N}_2 \perp \vec{N}_2$$

If it is also perpendicular to $\vec{N}_3$, then it gives a non-trivial solution.

General case: $AX = B$

$$\text{If } \det(A) \neq 0 \Rightarrow X = A^{-1}B \quad (\text{unique solution})$$

$$\text{If } \det(A) = 0 \Rightarrow \text{either no solution or infinitely many solutions}$$

Equation of Planes

We have touched on equations of planes previously. Here we fill in some of the details.

Planes in Point–Normal Form

The basic data which determines a plane is:

- A point P_0 in the plane
- A vector $\vec{N}$ orthogonal to the plane

Suppose we want the equation of a plane and we have:

$$P_0 = (x_0, y_0, z_0)$$

and a normal vector

$$\vec{N} = (a, b, c)$$

Let $P = (x, y, z)$ be an arbitrary point in the plane.

Then the vector $\overrightarrow{P_0P}$ lies in the plane and is orthogonal to $\vec{N}$. Therefore,

$$\vec{N} \cdot \overrightarrow{P_0P} = 0$$

$$\iff \langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

$$\iff a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Example 2: Find the plane containing the points

$$P_1 = (1, 2, 3), \quad P_2 = (0, 0, 3), \quad P_3 = (2, 5, 5).$$

Solution:

Vectors in the plane are

$$\overrightarrow{P_1P_2} = (-1, -2, 0), \quad \overrightarrow{P_1P_3} = (1, 3, 2).$$

A normal vector is their cross product:

$$\begin{aligned}\vec{N} = \vec{P_1P_2} \times \vec{P_1P_3} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -2 & 0 \\ 1 & 3 & 2 \end{vmatrix} \\ &= 4\mathbf{i} - 2\mathbf{j} - \mathbf{k}.\end{aligned}$$

Thus, a normal vector is

$$\vec{N} = (4, -2, -1).$$

Using point-normal form (with point P_1), the equation of the plane is

$$-4(x - 1) + 2(y - 2) - (z - 3) = 0$$

or equivalently,

$$-4x + 2y - z = -3.$$

Example 3: Find the plane with normal $\vec{N} = \mathbf{k}$ containing the point $(0, 0, 3)$.

Solution:

The equation of the plane is

$$\langle 0, 0, 1 \rangle \cdot \langle x, y, z - 3 \rangle = 0$$

which implies

$$z = 3.$$

Example 4: Find the plane with x , y , and z intercepts a , b , and c .

Solution:

Instead of using the previous method, we use a shortcut that works when the intercepts are known.

The intercepts are

$$(a, 0, 0), \quad (0, b, 0), \quad (0, 0, c).$$

Hence, the equation of the plane is

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$$

You can easily check that each of the given points is on the plane. For completeness, we will do this using the more general method of Example 2.

The three points give us two vectors in the plane

$$\langle -a, b, 0 \rangle \quad \text{and} \quad \langle a, 0, c \rangle.$$

Thus,

$$\vec{N} = \langle -a, b, 0 \rangle \times \langle a, 0, c \rangle = \langle bc, ac, ab \rangle.$$

Using the point-normal form,

$$bc(x - a) + ac(y - 0) + ab(z - 0) = 0.$$

Expanding,

$$bcx + acy + abz = abc.$$

Dividing by abc , we obtain

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$$

Lines of the Plane

When we are at it, let us look at two ways to write the equation of a line in the xy -plane.

Slope-intercept form:

Given the slope m and the y -intercept b , the equation of a line can be written as

$$y = mx + b.$$

Point-normal form:

We can also use the point-normal form to find the equation of a line. Given a point (x_0, y_0) on the line and a vector (a, b) normal to the line, the equation of the line is

$$a(x - x_0) + b(y - y_0) = 0.$$